Homework #2

(1) Compute the discrete-time equivalent system if

$$h(t) = \delta(t - \tau_0) + \alpha\delta(t - \tau_1)$$

(2) Find the power spectrum for a zero-mean WSS random process x(t) with an exponential correlation function with

$$R_{xx}(t) = e^{-2\beta|t|} \qquad t \in (-\infty, \infty)$$

(3) Suppose that a signal has to propagate over a distance of $100m$ to arrive at the receiver and undergoes an attenuation of 0.1. The baseband signal bandwidth is $5MHz$ and the carrier frequency is $f_c = 2GHz$.

a. Find a linear time invariant model $h_c(t)$ for this channel
b. Find the passband channel $h_p(t)$
c. Find the baseband equivalent channel $h(t)$
d. Find the discrete-time equivalent channel $h[n]$

(4) Perform the operations to obtain a zero mean normalized constellation from

$$C = \{j, \frac{1-j}{\sqrt{2}}, -j, \frac{-1-j}{\sqrt{2}}\}$$